KNOWLEDGE CHECKS

THE CLIENT-SERVER PARADIGM.

Which of the characteristics below are associated with a client-server approach to structuring network applications (as opposed to a P2P approach)?

- [x] There is a server with a well known server IP address.
- [ ] There is *not* a server that is always on.
- [x] There is a server that is always on.
- [x] HTTP uses this application structure.
- [ ] A process requests service from those it contacts and will provide service to processes that contact it.

That's Correct!

CHECK →

1/4

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KNOWLEDGE CHECKS

THE PEER-TO-PEER (P2P) PARADIGM.

Which of the characteristics below are associated with a P2P approach to structuring network applications (as opposed to a client-server approach)?

- [ ] HTTP uses this application structure.
- [x] A process requests service from those it contacts and will provide service to processes that contact it.
- [x] There is *not* a server that is always on.
- [ ] There is a server that is always on.
- [ ] There is a server with a well known server IP address.

That's Correct!

← CHECK →

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UDP SERVICE.

When an application uses a UDP socket, what transport services are provided to the application by UDP? Check all that apply.

- [ ] *Congestion control.* The service will control senders so that the senders do not collectively send more data than links in the network can handle.
- [ ] *Flow Control.* The provided service will ensure that the sender does not send so fast as to overflow receiver buffers.
- [ ] *Loss-free data transfer.* The service will reliably transfer all data to the receiver, recovering from packets dropped in the network due to router buffer overflow.
- [x] *Best effort service.* The service will make a best effort to deliver data to the destination but makes no guarantees that any particular segment of data will actually get there.
- [ ] *Real-time delivery.* The service will guarantee that data will be delivered to the receiver within a specified time bound.
- [ ] *Throughput guarantee.* The socket can be configured to provide a minimum throughput guarantee between sender and receiver.

That's Correct!

← CHECK →

3/4

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TCP SERVICE.

When an application uses a TCP socket, what transport services are provided to the application by TCP? Check all that apply.

- [x] *Congestion control.* The service will control senders so that the senders do not collectively send more data than links in the network can handle.
- [x] *Loss-free data transfer.* The service will reliably transfer all data to the receiver, recovering from packets dropped in the network due to router buffer overflow.
- [ ] *Throughput guarantee.* The socket can be configured to provide a minimum throughput guarantee between sender and receiver.
- [ ] *Best effort service.* The service will make a best effort to deliver data to the destination but makes no guarantees that any particular segment of data will actually get there.
- [x] *Flow Control.* The provided service will ensure that the sender does not send so fast as to overflow receiver buffers.
- [ ] *Real-time delivery.* The service will guarantee that data will be delivered to the receiver within a specified time bound.

That's Correct!

← CHECK

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KNOWLEDGE CHECKS

"HTTP IS STATELESS."

What do we mean when we say "HTTP is stateless"? In answering this question, assume that cookies are not used. Check all answers that apply.

- [ ] An HTTP client does not remember the identities of the servers with which it has interacted.
- [ ] The HTTP protocol is not licensed in any country.
- [x] An HTTP *server* does not remember anything about what happened during earlier steps in interacting with this HTTP client.
- [ ] We say this when an HTTP server is not operational.
- [ ] An HTTP *client* does not remember anything about what happened during earlier steps in interacting with any HTTP server.

That's Correct!

CHECK →

1/15

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HOME CHAPTER 2 - SECTION 2

KNOWLEDGE CHECKS

HTTP COOKIES.

What is an HTTP cookie used for?

- (selected) A cookie is a code used by a server, carried on a client's HTTP request, to access information the server had earlier stored about an earlier interaction with this Web *browser*. [Think about the distinction between a *browser* and a *person*.]
- A cookie is a code used by a client to authenticate a person's identity to an HTTP server.
- A cookie is used to spoof client identity to an HTTP server.
- A cookies is a code used by a server, carried on a client's HTTP request, to access information the server had earlier stored about an earlier interaction with this *person*. [Think about the distinction between a *browser* and a *person*.]
- Like dessert, cookies are used at the end of a transaction, to indicate the end of the transaction.

That's Correct!

← CHECK →

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KNOWLEDGE CHECKS

THE HTTP GET.

What is the purpose of the HTTP GET message?

- ○ The HTTP GET request message is sent by a web server to a web client to get the next request from the web client.
- ◉ The HTTP GET request message is used by a web client to request a web server to send the requested object from the server to the client.
- ○ The HTTP GET request message is sent by a web server to a web client to get the identity of the web client.
- ○ The HTTP GET request message is used by a web client to post an object on a web server.

That's Correct!

← CHECK →

3/15

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KNOWLEDGE CHECKS

CONDITIONAL HTTP GET.

What is the purpose of the conditional HTTP GET request message?

- (selected) To allow a server to only send the requested object to the client if this object has changed since the server last sent this object to the client.
- To allow a server to only send the requested object to the client if the client is authorized to received that object.
- To allow a server to only send the requested object to the client if the client has never requested that object before.
- To allow a server to only send the requested object to the client if the server is not overloaded.

That's Correct!

← CHECK →

4/15

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Suppose a client is sending an HTTP GET request message to a web server, gaia.cs.umass.edu. Suppose the client-to-server HTTP GET message is the following:

```
GET /kurose_ross_sandbox/interactive/quotation2.htm HTTP/1.1
Host: gaia.cs.umass.edu
Accept: text/plain, text/html, text/xml, image/jpeg, image/gif, audio/mpeg, audio/mp4, video/wmv, video/mp4,
Accept-Language: en-us, en-gb;q=0.1, en;q=0.7, fr, fr-ch, da, de, fi
If-Modified-Since: Wed, 09 Sep 2020 16:06:01 -0700
User Agent: Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/535.11 (KHTML, like Gecko) Chrome/17.0.963.56 Safari/535.11
```

What version of HTTP is the client using?

[Note: you can find additional questions similar to this here.]

- ○ 1
- ◉ 1.1
- ○ 2.1
- ○ 2

That's Correct!

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What is the language in which the client would least prefer to get a response? [You may have to search around the Web a bit to answer this.]

[Note: you can find additional questions similar to this here.]

- Mandarin
- Finnish
- United Kingdom English
- Farsi
- French
- Hindi
- US English
- Spanish

That's Correct!

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Again, suppose a client is sending an HTTP GET request message to a web server, gaia.cs.umass.edu. Suppose the client-to-server HTTP GET message is the following (same as in previous problem):

```
GET /kurose_ross_sandbox/interactive/quotation2.htm HTTP/1.1
Host: gaia.cs.umass.edu
Accept: text/plain, text/html, text/xml, image/jpeg, image/gif, audio/mpeg, audio/mp4, video/wmv, video/mp4,
Accept-Language: en-us, en-gb;q=0.1, en;q=0.7, fr, fr-ch, da, de, fi
If-Modified-Since: Wed, 09 Sep 2020 16:06:01 -0700
User Agent: Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/535.11 (KHTML, like Gecko) Chrome/17.0.963.56 Safari/535.11
```

Does the client have a cached copy of the object being requested?

[Note: you can find additional questions similar to this here.]

- There's not enough information in the header to answer this question.
- Yes, because this is a conditional GET, as evidenced by the If-Modified-Since field.
- Yes, because HTTP 1.1 is being used.
- No, because a client would not request an object if it had that object in its cache.

That's Correct!

← CHECK →

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Suppose now the server sends the following HTTP response message the client:

HTTP/1.0 200 OK
Date: Wed, 09 Sep 2020 23:46:21 +0000
Server: Apache/2.2.3 (CentOS)
Last-Modified: Wed, 09 Sep 2020 23:51:41 +0000
ETag:17dc6-a5c-bf716880.
Content-Length: 418
Connection: Close
Content-type: image/html

Will the web server close the TCP connection after sending this message?

[Note: you can find more questions like this one here.]

- There's not enough information in the response message to answer this question.
- No, the server will leave the connection open as a persistent HTTP connection.
- Yes, the server will close this connection because version 1.0 of HTTP is being used, and TCP connections do not stay open persistently.
- Yes, because the HTTP response indicated that only one object was requested in the HTTP GET request.

That's Correct!

← CHECK →

KNOWLEDGE CHECKS

WHY WEB CACHING?

Which of the following are advantages of using a web cache? Sselect one or more answers.

- [x] Caching uses less bandwidth coming into an institutional network where the client is located, if the cache is also located in that institutional network.
- [ ] Overall, caching requires fewer devices/hosts to satisfy a web request, thus saving on server/cache costs.
- [ ] Caching allows an origin server to more carefully track which clients are requesting and receiving which web objects.
- [x] Caching generally provides for a faster page load time at the client, if the web cache is in the client's institutional network, because the page is loaded from the nearby cache rather than from the distant server.

That's Correct!

← CHECK →

9/15

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KNOWLEDGE CHECKS

HTTP/2 VERSUS HTTP/1.1.

Which of the following are changes between HTTP 1.1 and HTTP/2? Note: select one or more answers.

- [ ] HTTP/2 provides enhanced security by using transport layer security (TLS).
- [x] HTTP/2 allows objects in a persistent connection to be sent in a client-specified priority order.
- [x] HTTP/2 allows a large object to be broken down into smaller pieces, and the transmission of those pieces to be interleaved with transmission other smaller objects, thus preventing a large object from forcing many smaller objects to wait their turn for transmission.
- [ ] HTTP/2 has many new HTTP methods and status codes.

That's Correct!

← CHECK →

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KNOWLEDGE CHECKS

WHAT'S IN AN HTTP REPLY?

Which of the following pieces of information will appear in a server's application-level HTTP reply message? (Check all that apply.)

- [ ] A sequence number
- [ ] The server's IP address
- [x] A response code
- [ ] The name of the Web server (e.g., gaia.cs.umass.edu)
- [ ] A checksum
- [x] A response phrase associated with a response code

That's Correct!

← CHECK →

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KNOWLEDGE CHECKS

IF-MODIFIED-SINCE.

What is the purpose of the *If-Modified-Since* field in a HTTP GET request message

- (selected) To indicate to the server that the client has cached this object from a previous GET, and the time it was cached.
- To indicate to the server that the client wishes to receive this object, and the time until which it will cache the returned object in the browser's cache.
- To allow the server to indicate to the client that it (the client) should cache this object.
- To indicate to the server that the server should replace this named object with the new version of the object attached to the GET, if the object has not been modified since the specified time
- To inform the HTTP cache that it (the cache) should retrieve the full object from the server, and then cache it until the specified time.

That's Correct!

← CHECK →

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KNOWLEDGE CHECKS

COOKIES.

What is the purpose of a cookie value in the HTTP GET request?

- The cookie value encodes a default set of preferences that the user has previously specified for this web site.
- The cookie value is an encoding of a user email address associated with the GET request.
- The cookie value itself doesn't mean anything. It is just a value that was returned by a web server to this client during an earlier interaction.
- The cookie value encodes the format of the reply preferred by the client in the response to this GET request.
- The cookie value indicates whether the user wants to use HTTP/1, HTTP/1.1, or HTTP/2 for this GET request.

That's Correct!

← CHECK →

13/15

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HTTP GET (EVEN MORE).

Suppose a client is sending an HTTP GET message to a web server, gaia.cs.umass.edu. Suppose the client-to-server HTTP GET message is the following:

```
GET /kurose_ross_sandbox/interactive/quotation2.htm HTTP/1.1
Host: gaia.cs.umass.edu
Accept: text/plain, text/html, text/xml, image/jpeg, image/gif, audio/mpeg, audio/mp4, video/wmv, video/mp4,
Accept-Language: en-us, en-gb;q=0.1, en;q=0.7, fr, fr-ch, da, de, fi
If-Modified-Since: Wed, 09 Sep 2020 16:06:01 -0700
User Agent: Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/535.11 (KHTML, like Gecko) Chrome/17.0.963.56 Safari/535.11
```

Does the client have a cached copy of the object being requested?

- There's not enough information to answer this question.
- No, because the client would not request an object if it were cached.
- Yes, because this is a conditional GET.

That's Correct!

← CHECK →

WHAT HAPPENS AFTER AN HTTP REPLY?

Suppose an HTTP server sends the following HTTP response message a client:

HTTP/1.0 200 OK
Date: Wed, 09 Sep 2020 23:46:21 +0000
Server: Apache/2.2.3 (CentOS)
Last-Modified: Wed, 09 Sep 2020 23:51:41 +0000
ETag:17dc6-a5c-bf716880.
Content-Length: 418
Connection: Close
Content-type: image/html

Will the web server close the TCP connection after sending this message?

- ○ There's not enough information to answer this question.
- ○ No, this is a persistent connection, and so the server will keep the TCP connection open.
- ◉ Yes, because this is HTTP 1.0

That's Correct!

← CHECK

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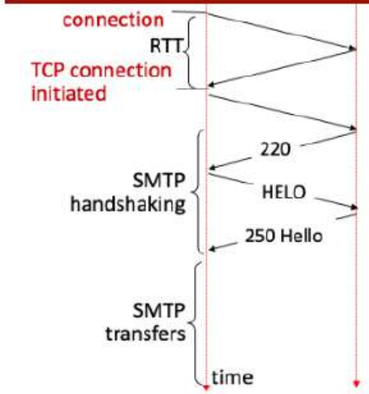

- ○ 2
- ○ 0
- ◉ 3
- ○ 2.5
- ○ 1

That's Correct!

CHECK →

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COMPARING AND CONTRASTING HTTP AND SMTP.

Which of the following characteristics apply to HTTP only (and do *not* apply to SMTP)? Note: check one or more of the characteristics below.

- [x] Uses a blank line (CRLF) to indicate end of request header.
- [x] Operates mostly as a “client pull” protocol.
- [ ] Operates mostly as a “client push” protocol.
- [ ] Uses server port 25.
- [ ] Uses CRLF.CRLF to indicate end of message.
- [ ] Has ASCII command/response interaction, status codes.
- [x] Uses server port 80.
- [ ] Is able to use a persistent TCP connection to transfer multiple objects.

That's Correct!

← CHECK →

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COMPARING AND CONTRASTING HTTP AND SMTP (2).

Which of the following characteristics apply to SMTP only (and do *not* apply to HTTP)? Note: check one or more of the characteristics below.

- [ ] Uses a blank line (CRLF) to indicate end of request header.
- [ ] Is able to use a persistent TCP connection to transfer multiple objects.
- [ ] Has ASCII command/response interaction, status codes.
- [ ] Operates mostly as a "client pull" protocol.
- [x] Operates mostly as a "client push" protocol.
- [ ] Uses server port 80.
- [x] Uses server port 25.
- [x] Uses CRLF.CRLF to indicate end of message.

That's Correct!

← CHECK →

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KNOWLEDGE CHECKS

COMPARING AND CONTRASTING HTTP AND SMTP (3).

Which of the following characteristics apply to both HTTP and SMTP? Note: check one or more of the characteristics below.

- [ ] Uses CRLF.CRLF to indicate end of message.
- [ ] Operates mostly as a "client push" protocol.
- [ ] Operates mostly as a "client pull" protocol.
- [x] Has ASCII command/response interaction, status codes.
- [x] Is able to use a persistent TCP connection to transfer multiple objects.
- [ ] Uses a blank line (CRLF) to indicate end of request header.

That's Correct!

← CHECK →

Match the functionality of a protocol with the name of a the email protocol (if any) that implements that functionality.

QUESTION LIST:

Pushes email from a mail client to a mail server.

A

Pulls mail from one mail server to another mail server.

C

Pulls email to a mail client from a mail server.

B

ANSWER LIST:

A. SMTP

B. IMAP

C. Neither SMTP nor IMAP does this.

That's Correct!

← CHECK

5/5

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HOME

CHAPTER 2 - SECTION 4

Provides authoritative hostname to IP mappings for organization's named hosts.

B

Replies to DNS query by local host, by contacting other DNS servers to answer the query.

A

Responsible for a domain (e.g., *.com, *.edu); knows how to contact authoritative name servers.

C

Highest level of the DNS hierarchy, knows how to reach servers responsible for a given domain (e.g., *.com, *.edu).

D

A. Local DNS server

B. Authoritative DNS server

C. Top Level Domain (TLD) servers

D. DNS root servers

That's Correct!

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KNOWLEDGE CHECKS

WHY DOES THE DNS PERFORM CACHING?

What is the value of caching in the local DNS name server? Check all that apply.

- [ ] DNS caching provides the ability to serve as authoritative name server for multiple organizations.
- [x] DNS caching provides for faster replies, if the reply to the query is found in the cache.
- [ ] DNS caching provides prioritized access to the root servers, since the DNS request is from a local DNS cache.
- [x] DNS caching results in less load elsewhere in DNS, when the reply to a query is found in the local cache.

That's Correct!

← CHECK →

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HOME

CHAPTER 2 - SECTION 4

KNOWLEDGE CHECKS

WHAT'S IN THE DNS TYPE A RESOURCE RECORD?

What information does the type "A" resource record hold in the DNS database? Check all that apply.

- [ ] A name and the name of the SMTP server associated with that name.
- [ ] A domain name and the name of the authoritative name server for that domain.
- [x] A hostname and an IP address.
- [ ] An alias name and a true name for a server.

That's Correct!

← CHECK →

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Consider the following DNS requests, made by the local host at the given times:

- *t=0*, the local host requests that the name gaia.cs.umass.edu be resolved to an IP address.
- *t=1*, the local host requests that the name icann.org be resolved to an IP address.
- *t=5*, the local host requests that the name cs.umd.edu be resolved to an IP address. (Hint: be careful!)
- *t=10*, the local host ***again*** requests that the name gaia.cs.umass.edu be resolved to an IP address.
- *t=12*, the local host requests that the name cs.mit.edu be resolved to an IP address.
- *t=30*, the local host ***again*** requests that the name gaia.cs.umass.edu be resolved to an IP address. (Hint: be careful!)

Which of the requests require 8 time units to be resolved?

- [ ] The request at *t=12*.
- [x] The request at *t=0*.
- [ ] The request at *t=5*.
- [x] The request at *t=1*.
- [x] The request at *t=30*.
- [ ] The request at *t=10*.

That's Correct!

← CHECK →

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Consider the following DNS requests, made by the local host at the given times:

- *t=0*, the local host requests that the name gaia.cs.umass.edu be resolved to an IP address.
- *t=1*, the local host requests that the name icann.org be resolved to an IP address.
- *t=5*, the local host requests that the name cs.umd.edu be resolved to an IP address. (Hint: be careful!)
- *t=10*, the local host *again* requests that the name gaia.cs.umass.edu be resolved to an IP address.
- *t=12*, the local host requests that the name cs.mit.edu be resolved to an IP address.
- *t=30*, the local host *again* requests that the name gaia.cs.umass.edu be resolved to an IP address. (Hint: be careful!)

Which of the requests require 6 time units to be resolved?

- [ ] The request at *t=10*.
- [x] The request at *t=5*.
- [x] The request at *t=12*.
- [ ] The request at *t=30*.
- [ ] The request at *t=0*.
- [ ] The request at *t=1*.

That's Correct!

← CHECK →

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Knowledge Checks

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HOME

CHAPTER 2 - SECTION 4

Consider the following DNS requests, made by the local host at the given times:

- *t=0*, the local host requests that the name gaia.cs.umass.edu be resolved to an IP address.
- *t=1*, the local host requests that the name icann.org be resolved to an IP address.
- *t=5*, the local host requests that the name cs.umd.edu be resolved to an IP address. (Hint: be careful!)
- *t=10*, the local host *again* requests that the name gaia.cs.umass.edu be resolved to an IP address.
- *t=12*, the local host requests that the name cs.mit.edu be resolved to an IP address.
- *t=30*, the local host *again* requests that the name gaia.cs.umass.edu be resolved to an IP address. (Hint: be careful!)

Which of the requests require 2 time units to be resolved?

- [ ] The request at *t=30*.
- [ ] The request at *t=5*.
- [ ] The request at *t=12*.
- [x] The request at *t=10*.
- [ ] The request at *t=1*.
- [ ] The request at *t=0*.

That's Correct!

← CHECK →

6/9

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KNOWLEDGE CHECKS

THE LOCAL DNS SERVER.

Check all of the phrases below that state a true property of a *local* DNS server.

- [ ] The local DNS server is only contacted by a local host if that local host is unable to resolve a name via iterative or recursive queries into the DNS hierarchy.
- [x] The local DNS server can decrease the name-to-IP-address resolution time experienced by a querying local host over the case when a DNS is resolved via querying into the DNS hierarchy.
- [x] The local DNS server record for a remote host is sometimes different from that of the authoritative server for that host.
- [ ] The local DNS server holds hostname-to-IP translation records, but not other DNS records such as MX records.

That's Correct!

← CHECK →

7/9

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KNOWLEDGE CHECKS

THE DNS AUTHORITATIVE NAME SERVER.

What is the role of an authoritative name server in the DNS? (Check all that apply)

- [ ] It provides a list of TLD servers that can be queried to find the IP address of the DNS server that can provide the definitive answer to this query.
- [ ] It is a local (to the querying host) server that caches name-to-IP address translation pairs, so it can answer authoritatively and can do so quickly.
- [ ] It provides the IP address of the DNS server that can provide the definitive answer to the query.
- [x] It provides the definitive answer to the query with respect to a name in the authoritative name server's domain.

That's Correct!

← CHECK →

8/9

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KNOWLEDGE CHECKS

DNS AND HTTP CACHING.

We learned that in HTTP web browser caching, HTTP local web server caching, and in local DNS caching, that a user benefits (e.g., shorter delays over the case of no caching) from finding a local/nearby copy of a requested item. In which of the following forms of caching does a user benefit from its not only from its own recent requests (and cached replies) *but also from recent requests made from other users*?

- [x] HTTP local web caching
- [x] Local DNS server caching
- [ ] HTTP browser caching

That's Correct!

← CHECK

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CHAPTER 2 - SECTION 7

UDP SOCKETS:

Which of the following characteristics below are associated with a UDP socket? Check one or more that apply.

- [ ] socket(AF_INET, SOCK_STREAM) creates this type of socket
- [x] the application must explicitly specify the IP destination address and port number for each group of bytes written into a socket
- [ ] provides reliable, in-order byte-stream transfer (a "pipe"), from client to server
- [x] data from different clients can be received on the same socket
- [ ] when contacted, the server will create a new server-side socket to communicate with that client
- [ ] a server can perform an accept() on this type of socket
- [x] provides unreliable transfer of a groups of bytes ("a datagram"), from client to server
- [x] socket(AF_INET, SOCK_DGRAM) creates this type of socket

That's Correct!

CHECK →

1/5

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Which of the following characteristics below are associated with a TCP socket? Check one or more that apply.

- [x] socket(AF_INET, SOCK_STREAM) creates this type of socket
- [ ] data from different clients can be received on the same socket
- [x] a server can perform an accept() on this type of socket
- [x] provides reliable, in-order byte-stream transfer (a "pipe"), from client to server
- [ ] socket(AF_INET, SOCK_DGRAM) creates this type of socket
- [x] when contacted, the server will create a new server-side socket to communicate with that client
- [ ] provides unreliable transfer of a group of bytes (a "datagram"), from client to server
- [ ] the application must explicitly specify the IP destination address and port number for each group of bytes written into a socket

That's Correct!

← CHECK →

2/5

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SERVER REPLY (UDP).

How does the networked application running on a server know the client IP address and the port number to reply to in response to a received datagram?

- The server will query the DNS to learn the IP address of the client.
- As the result of performing the accept() statement, the server has created a new socket that is bound to that specific client, and so sending into this new socket (without explicitly specifying the client IP address and port number) is sufficient to ensure that the sent data will be addressed to the correct client.
- The server will know the port number being used by the client since all services have a well-known port number.
- The application code at the server determines client IP address and port # from the initial segment sent by client, and must explicitly specify these values when sending into a socket back to that client.

That's Correct!

← CHECK →

3/5

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KNOWLEDGE CHECKS

HOW MANY SOCKETS?

Suppose a Web server has *five* ongoing connections that use TCP receiver port 80, and assume there are no other TCP connections (open or being opened or closed) at that server. How many TCP sockets are in use at this server?

- ○ 5
- ◉ 6
- ○ 1
- ○ 4

That's Correct!

← CHECK →

4/5

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Knowledge Checks

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CHAPTER 2 - SECTION 7

KNOWLEDGE CHECKS

SOCKET CONNECT().

What happens when a socket connect() procedure is called/invoked?

- This causes the server to reach out to a TCP client to establish a connection between that client and the server.
- This procedure creates a new socket at the client, and connects that socket to the specified server.
- This causes the client to reach out to a TCP server to establish a connection between that client and the server. If there is already one or more servers on this connection, this new server will also be added to this connection.

That's Correct!

← CHECK

5/5

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